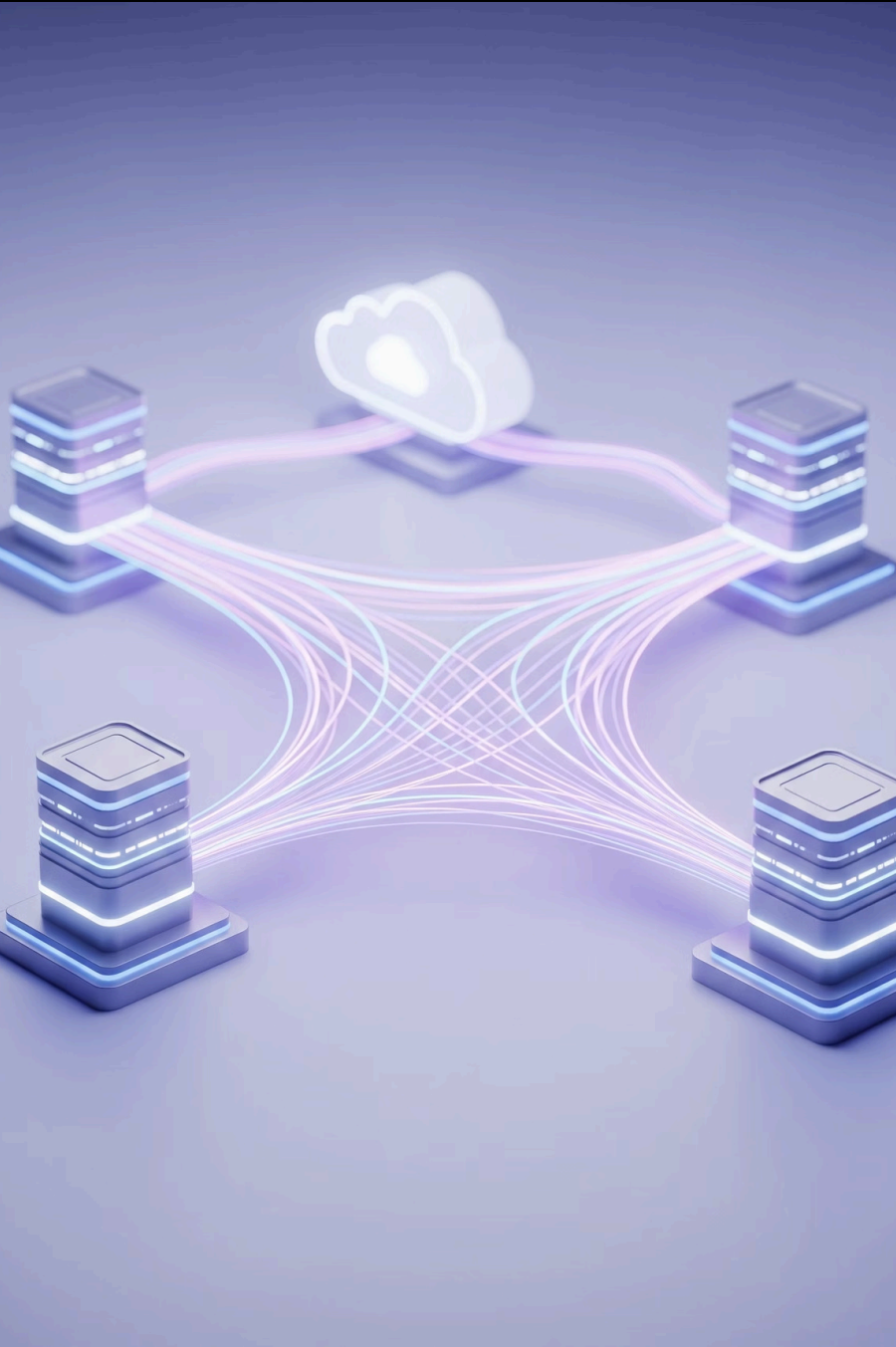


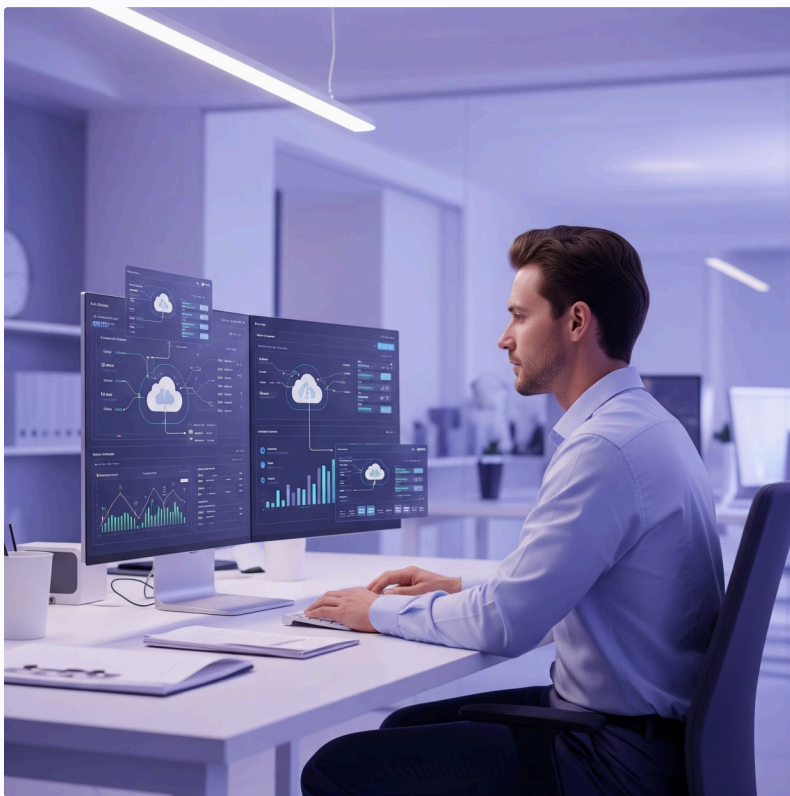


Data Replication Across Cloud Providers

Keeping AI data consistent, available, and resilient

A technical guide for building robust multi-cloud AI systems

Why Replicate Data?



Multi-cloud AI workloads require **consistent data access** across environments

Disaster Recovery

Maintain business continuity across regions/providers during outages

Low-Latency Access

Position data closer to compute resources and end users

Compliance

Meet data residency requirements in regulated industries

Cross-Cloud Pipelines

Enable seamless training/inference across cloud boundaries

Key Challenges

Effective multi-cloud data replication requires navigating several technical hurdles

Data Gravity

Large AI datasets (TBs/PBs) create significant inertia and are expensive to move across cloud boundaries

Latency & Consistency

Fundamental trade-offs between real-time consistency and performance, especially at scale

Egress Fees

Cross-cloud data transfer costs can exceed storage costs by orders of magnitude

Schema Drift

Format and schema inconsistencies between cloud systems can corrupt data integrity

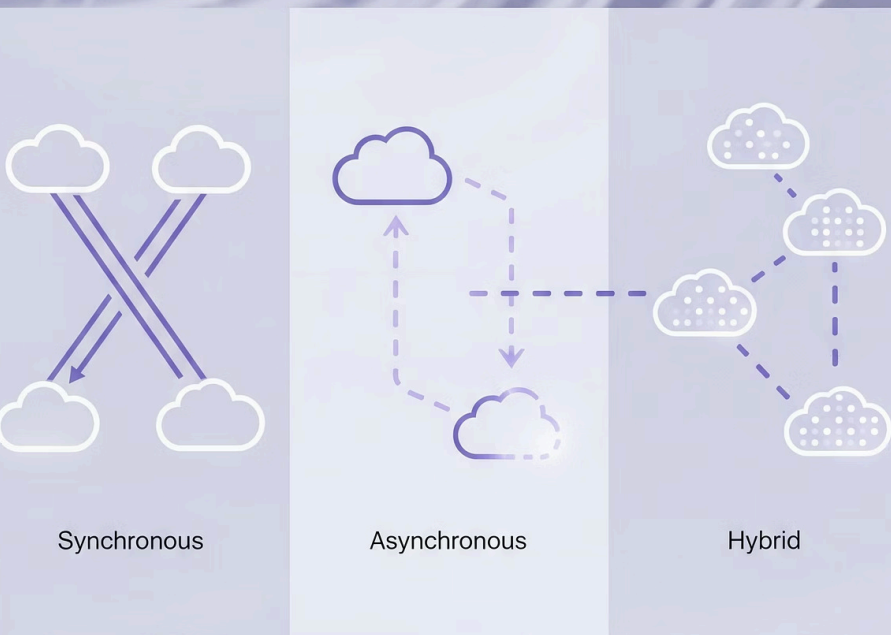
Security

Maintaining encryption and access controls during transit between environments



Cloud providers typically charge for **outbound** data transfers but not for **inbound** transfers

Replication Models



Synchronous Replication

- Real-time consistency between sources
- Transactions commit only after all replicas confirm
- Higher latency and operational costs
- Best for critical transaction data

Asynchronous Replication

- Eventual consistency model
- Primary system commits, secondaries update later
- Better performance and lower costs
- Acceptable for most AI training data

Hybrid Replication

- Sync for metadata and critical configs
- Async for bulk data and large assets
- Balances performance and consistency
- Ideal for most ML production systems

Storage Replication Options

Each cloud storage layer requires specific replication approaches



Object Storage

S3 ⇄ GCS ⇄ Azure Blob

- CloudSync, rclone, storage gateway
- Native bucket replication
- CDN edge caching for static assets



Databases

Cross-cloud managed DB options

- Cloud Spanner multi-region
- Aurora Global Database
- Cosmos DB multi-region write
- MongoDB Atlas multi-cloud clusters



Data Lakes

Portable open formats

- Apache Iceberg
- Delta Lake
- Apache Hudi
- Log-based incremental replication



Streaming

Real-time data movement

- Kafka MirrorMaker 2
- Pub/Sub ⇄ Kafka bridges
- Confluent Cloud multi-cloud

Feature Store & Vector DB Replication

ML-Specific Data Systems

Feature Stores

Feast, Tecton with multi-cloud storage backends

- Consistent feature values across clouds
- Offline/online store replication

Vector Databases

Pinecone, Milvus, Weaviate with active-active replication

- Embedding consistency for distributed inference
- Read replicas for vector search



Critical for ensuring **consistent model predictions** across cloud environments

Example: Kafka MirrorMaker 2



connectors:

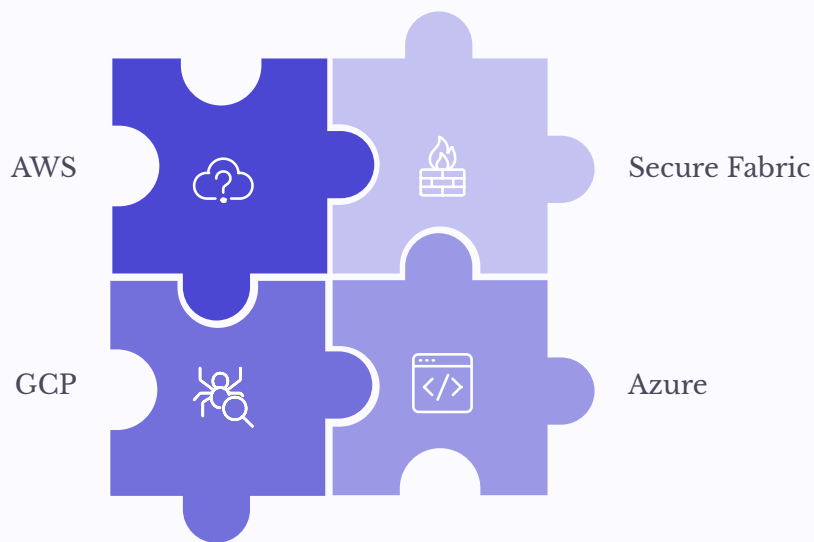
```
name: mm2-connector
source.cluster.alias: aws-cluster
target.cluster.alias: gcp-cluster
replication.policy.class:
  org.apache.kafka.connect.mirror.DefaultReplicationPolicy
```

Streams AI features/logs from AWS → GCP

Key Benefits

- Enables **cross-cloud event consistency**
- Active-active multi-cluster replication
- Preserves offset mapping between clouds
- Supports topic renaming policies

Network & Security Considerations



Private Interconnects

AWS Direct Connect ↔ Google Cloud Interconnect ↔
Azure ExpressRoute

- Reduced latency and egress costs
- Isolated from public internet



Encryption Requirements

End-to-end protection for sensitive AI data

- TLS/mTLS for all replication traffic
- Encrypt data at rest in both providers
- Key management across clouds (KMS, Key Vault)



Monitoring & Alerting

Comprehensive visibility across boundaries

- Track replication lag metrics
- Alert on failed transfers
- Audit data consistency regularly

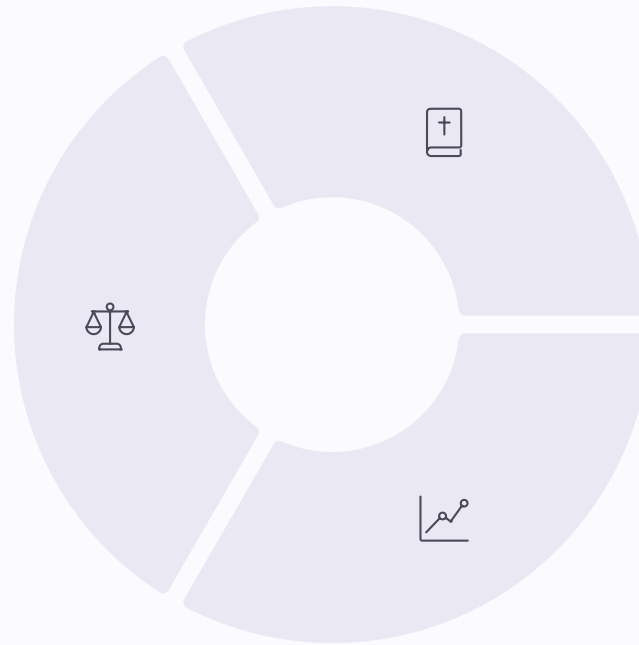
Trade-Offs

Balancing competing requirements for multi-cloud replication

Performance vs. Cost

Synchronous replication provides stronger consistency but significantly increases:

- Network costs (egress fees)
- Latency for writes
- Operational complexity



Compliance vs. Agility

Regulatory requirements can limit replication options:

- Data sovereignty restrictions
- Required local copies in some regions
- Audit trail requirements

Ops Complexity

Multi-cloud replication increases:

- Tool chain diversity
- Monitoring complexity
- Recovery procedure complexity

❏ Cloud architects must align replication strategies with specific AI workload requirements and SLAs

Best Practices



1 Use Portable Data Formats

Standardize on open formats like Parquet, Avro, and table formats (Iceberg, Delta) for data lakes and warehouses

2 Replicate Critical Subsets

Identify and prioritize high-value data - not everything needs to be replicated across all clouds

3 Monitor Replication Lag

Implement custom metrics and alerts for replication health and consistency measurements

4 Align With AI Workloads

Training data can tolerate async; inference may require sync replication for consistency

5 Test DR Regularly

Schedule quarterly recovery drills to validate cross-cloud resilience

Key Takeaways

Data replication is the backbone of multi-cloud AI success



Sync vs. Async

Choose replication strategy based on SLA requirements and workload type



Open Formats

Leverage portable data formats and managed replication services



Security First

Implement end-to-end encryption and comprehensive access controls



Monitor & Test

Continuous monitoring and regular disaster recovery drills are essential

Questions? Contact the Cloud Architecture Team